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ASSIGHNMENT # 01

SUBMITTED TO: MISS AMBREEN GUL

Question 1:

CODE:

```python
def caesar_encrypt(text, shift):  1 usage
    encrypted_text = ""

    for char in text:
        if char.isalpha():
            if char.isupper():
                encrypted_text += chr((ord(char) - 65 + shift) % 26 + 65)
            else:
                encrypted_text += chr((ord(char) - 97 + shift) % 26 + 97)
        else:
            encrypted_text += char
    return encrypted_text
def caesar_decrypt(ciphertext, shift):  1 usage
    decrypted_text = ""
    for char in ciphertext:
        if char.isalpha():
            if char.isupper():
                decrypted_text += chr((ord(char) - 65 - shift) % 26 + 65)
            else:
                decrypted_text += chr((ord(char) - 97 - shift) % 26 + 97)
        else:
            decrypted_text += char
    return decrypted_text
```

```python
    return decrypted_text
message = input("Enter your message: ")
shift = int(input("Enter shift value: "))
encrypted = caesar_encrypt(message, shift)
print("Encrypted Message:", encrypted)
decrypted = caesar_decrypt(encrypted, shift)
print("Decrypted Message:", decrypted)
```

OUTPUT:

```
C:\Users\zunai\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\zunai\PycharmProjects\PythonProject\..py
Enter your message: sabahameed
Enter shift value: 3
Encrypted Message: vdedkdphhg
Decrypted Message: sabahameed

Process finished with exit code 0
```

Simple Explanation of Caesar Cipher Code

This Python program implements **Caesar Cipher encryption and decryption**.

The program contains **two main functions**:

1. caesar_encrypt(text, shift)
2. caesar_decrypt(ciphertext, shift)

1. caesar_encrypt(text, shift)

This function is used to **encrypt the given message**.

- First, an empty string is created to store the encrypted text.
- The program reads the input message **character by character using a loop**.
- If the character is a letter:
  - It checks whether the letter is **uppercase or lowercase**.
  - The letter is **shifted forward** using the given shift value.
  - The shifted letter is added to the encrypted text.
- If the character is **not a letter** (such as space or special symbol), it is **added without any change**.

In this way, only letters are encrypted, while spaces and special characters remain unchanged.

2. caesar_decrypt(ciphertext, shift)

This function is used to **decrypt the encrypted message** and get back the original text.

- First, an empty string is created to store the decrypted text.
- The program reads the encrypted message **character by character**.
- If the character is a letter:
  - The shift value is **subtracted** to reverse the encryption.
  - The original character is recovered.

- If the character is **not a letter**, it remains unchanged.

This function correctly converts the encrypted text back into readable form.

3. Main Program Working

- The program takes a message and a shift value from the user.
- It calls the encryption function to encrypt the message.
- Then it calls the decryption function to decrypt the encrypted message.
- This confirms that the encryption and decryption process is working correctly.

Conclusion

This program correctly implements the **Caesar Cipher technique**.
It encrypts and decrypts text while:

- Maintaining uppercase and lowercase letters.